

ComicCraft - AI Comic Story Creator using Gemini Models

6. Gemini AI Integration

Technology Track: Generative AI with Google

1. Overview of Gemini Integration

ComicCraft relies on two Gemini capabilities working in sequence: the Gemini text generation model plans the story into panels and writes captions, and the Gemini image generation model creates the artwork for each panel based on those plans.

2. Text Generation Integration

2.1 Purpose

The Gemini text model receives the user's story idea along with an instruction to split it into a fixed number of panels, each described as a short scene plus a caption or dialogue line, returned in a structured (JSON) format.

2.2 Sample Prompt Structure

A typical instruction sent to the text model asks it to: read the story, divide it into N panels, describe the visual scene for each panel in one or two sentences, and write a short caption or line of dialogue for that panel — all formatted as a JSON list the backend can loop through.

3. Image Generation Integration

3.1 Purpose

For each panel's scene description returned by the text model, the Gemini image model generates the corresponding artwork, guided by both the scene description and a fixed style/character phrase that is reused across every panel in the same comic.

3.2 Maintaining Visual Consistency

- A single, detailed character/style description (appearance, color palette, art style) is defined once per comic session.
- That same description is appended to every panel's image prompt, so the character and setting look consistent from the first panel to the last.
- Only the scene-specific action or setting text changes between panel prompts.

4. Integration Flow

Step	Action
1	User story is sent to the Gemini text model with panel-count and formatting instructions.
2	Text model returns a structured list of scene descriptions and captions.
3	Each scene description, combined with the fixed style/character phrase, is sent to the Gemini image model.
4	Image model returns one artwork image per panel.
5	Backend combines all panel images and captions into the final comic strip.

5. Handling API Responses and Errors

- Text responses are parsed as JSON; malformed responses trigger a retry with a stricter formatting instruction.
- Image requests that fail or time out are retried once before showing the user a clear error message.
- All API calls are made server-side so that credentials are never exposed to the browser.

6. Safety Considerations

- Both text and image requests rely on Gemini's built-in content safety filtering.
- Prompts flagged as unsafe are blocked before reaching the image generation step.

7. Summary

By chaining Gemini's text and image generation together — with a shared style description carried across every panel — ComicCraft reliably converts a plain story idea into a coherent, illustrated comic strip.